



UNIVERSITÀ
DI TRENTO



Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY
OF NEURAL CIRCUITS

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Abstract

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Chapter 1

Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input neurons respond with different patterns of spike timings, while the overall firing rate remains roughly equal [1, 2, 3].

The sources of variability